

Occam's Worm

Progress report: what we have built and what the data says so far

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Summary

- **The pipeline is complete end to end.** It runs from data import and audit, through animal-held-out splits, scoring, baselines, a typed rule language with three agreeing simulators, fitting and a nested rule search, to locked evaluation. 14 of 16 tickets are done; the other two are waiting only on compute.
- **Stimulus responses are predictable in animals never seen.** On the primary task (T2a, 110 held-out animals), free per-pair response kernels beat “nothing happens” by +105 (95% CI +22 to +182) log-likelihood units per animal.
- **Most of that structure is shared.** One response shape per stimulated neuron (B1) is statistically indistinguishable from a free shape for every pair: -3 (95% CI -7 to +1). A bank of four shared shapes (B2-K4) matches the free table exactly: +0.01 (95% CI -0.17 to +0.21).
- **The wiring diagram carries about half of the signal.** A linear network on the anatomical wiring (B4) improves on “nothing happens” by +54 (95% CI +32 to +77), in 83% of animals. That is roughly half of what the free table gains, with 311 parameters instead of about 322,000.
- **Short rules reach the same regime.** A 150-bit hand-written rule (leaky integration plus chemical and gap-junction coupling) sits on the size/accuracy Pareto front in every fold tested. The free table still wins outright.
- **Neuropeptide routes add a small, consistent gain over a shuffled control.** Adding Ripoll-Sanchez 2023 peptide routes to that rule helps on all three folds tested, slightly more than an equally sized label-shuffled peptide graph. Only one shuffle has been run so far.
- **Two problems were caught and fixed:** a stimulation-light artifact in the recordings, and a bias in how the ridge penalty was chosen. Each was fixed before any confirmatory result.

Status. The full baseline benchmark has completed 1 of its 4 configurations (T2a without a stimulation-history term). The other three need about 15 more hours of compute on a machine that stays awake. The search results come from 3 of 113 outer folds and are exploratory.

1. The question

C. elegans is the only animal whose complete wiring diagram has been mapped several times. We use the Cook et al. (2019) hermaphrodite reconstruction: 302 neurons, about 3,700 chemical connections and about 1,100 gap-junction pairs. A wiring diagram is not a model, though. It does not say whether a synapse excites or inhibits, how quickly a neuron forgets its input, or how inputs combine.

The project asks whether there is a **short, local update rule** that every neuron runs and that turns the wiring diagram into a predictor of real causal responses. It must work in animals that played no part in choosing or fitting it. Held-out accuracy comes first; description length only breaks ties and defines a Pareto front of size against accuracy.

The data is the Randi et al. (2023) signal-propagation atlas. One neuron at a time is stimulated optogenetically while calcium activity is imaged across the head at 2 volumes per second. Each trial has 10 volumes before the pulse and 60 after. The import covers 113 wild-type recordings. In task **T2a** a model receives the stimulated neuron's own measured response as input; in **T2b** it receives only a nominal impulse and must also model the calcium indicator.

2. How results are protected from overfitting

- **Split by animal, never by trial.** The split was chosen from coverage tables before any model was scored, and frozen with a sha256 hash. Leave-one-animal-out (113 outer folds × 5 inner folds) covers 144 eligible stimulation targets, against 125 and 127 for 5- and 10-fold splits.
- **Nested evaluation.** Every choice (model family, ridge penalty, noise model, which rule wins) is made on inner folds of the training animals. The held-out animal is scored once.
- **Locks.** The search code receives training-only copies of the data. Each fold's choice is frozen with its own sha256 before scoring. A separate evaluator refuses three things: tampered selections, selections trained on held-out animals, and a second scoring of the same fold. Injected leaks are tested and fail.
- **Same yardstick for everything.** Every model predicts response kernels on the same 12-knot basis and is scored by the same correlated-noise likelihood. That is a Gaussian AR(1) with a per-trace scale $\sigma^2 = a^2s^2 + b^2$, where s is the trace's pre-stimulus noise; the noise is fitted on training residuals only.
- **Search volume is recorded.** Every attempted candidate is logged, including those rejected by structural gates. Budget curves compare the search order against random orders.

3. What has been built

Area	State
Data	Source registry and acquisition (OW-001); trial importer (OW-002); M0 audit, verdict go (OW-003); frozen animal-wise split (OW-004); annotation and connectome mapping, 302 neurons and 23 reconstructions (OW-013).
Scoring	Masked exact AR(1) likelihood, block scoring, a separately parameterized calcium-indicator stage, and an animal-level bootstrap (OW-005).
Baselines	B0–B3 kernel families and the B4 linear anatomical network, all in one nested benchmark (OW-006/007, implemented; full run in progress).
Rule language	WRL: a typed, unit-checked language with a C++26 compiler and canonical IR (OW-008), a reference interpreter (OW-009), and an exactly-once enumerator (OW-010).
Simulators	C++ reference, scalar Python and differentiable JAX implementations, agreeing to 4.4e-16 on the conformance suite (OW-009).
Fitting	Bounded transforms, multi-start L-BFGS-B, Hessian and bootstrap standard errors; synthetic parameters are recovered within 3 SE (OW-011).
Search	Training-only views, frozen selections, a locked evaluator, WRL candidates on the connectome, gates, budget curves (OW-012).
Downstream	Experiment planner and equivalence classes (OW-014), provenance-tracked reports (OW-015), and a closed-loop body interface (OW-016).

More than 420 automated tests cover these components. They include leakage tests, synthetic ground-truth recovery, and cross-implementation conformance. All passed at the last local run.

4. Result: the baseline benchmark (T2a, no history term)

All 113 outer folds of the first configuration are complete. 110 held-out animals were scored; 3 animals have no eligible traces. Lower NLL is better. The totals are sums of log-densities over about six million samples, so the differences that matter look small per sample.

Family	What it is	Held-out NLL	per sample	Params (median)	Ridge (mode)
B0	no response, drift only	-690,457	-0.1141	7	0
B1	one shape per target	-701,672	-0.1159	28,753	100
B1d	B1 + per-pair delays	-701,694	-0.1159	55,591	100
B2-K1	1 global shape	-701,415	-0.1159	26,857	100
B2-K2	bank of 2 shapes	-701,979	-0.1160	53,707	100
B2-K4	bank of 4 shapes	-702,022	-0.1160	107,407	100
B2-K8	bank of 8 shapes	-702,021	-0.1160	214,807	100
B3	free shape per pair	-702,020	-0.1160	322,063	100
B4-fixed	linear net on wiring, 3 params	-696,391	-0.1151	10	0.1
B4-learned	linear net, gain per neuron	-696,436	-0.1151	311	0.1
nested	family chosen per fold on inner folds	-702,672			

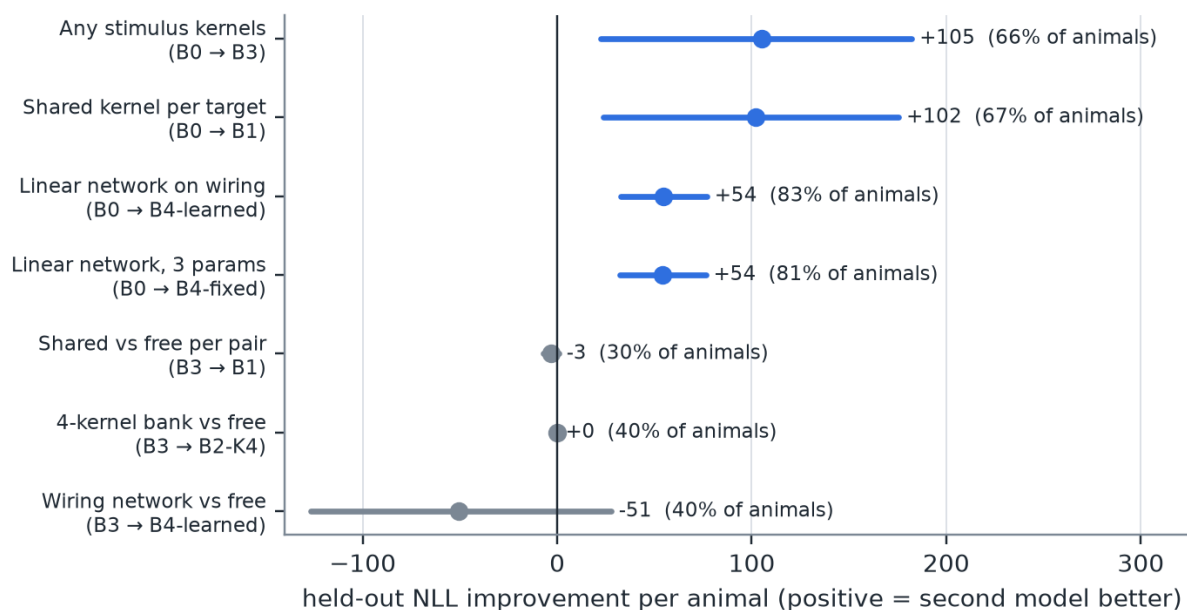


Figure 1. Paired held-out comparisons, one value per animal, with an animal-level bootstrap (2,000 replicates). Blue: the second model is better, CI excludes 0. Grey: CI includes 0.

What this says

- **Responses to stimulation are predictable across animals.** Every kernel family beats B0. B3 improves by 105 log-likelihood units per held-out animal, and 66% of animals favour it.
- **Pair-specific detail adds little.** B1, with one shape per stimulated neuron scaled per responder, is within noise of B3: -3 (95% CI -7 to +1). A bank of 4 shared shapes is equivalent to B3. Inner-fold selection picked B2-K4 in 85 of 110 folds, and that nested choice has a total held-out NLL 652 lower than B3's.

- **Anatomy alone explains about half.** B4 (linear dynamics on the Cook wiring, signs from Fenyves 2020) gains 54 over B0 and helps in 83% of animals. It trails B3 by 51 on average, with a wide CI (-127 to +28).
- **The noise model is adequate but not exact.** Central 50% intervals cover about 62% of held-out samples, and 95% intervals cover about 96%. Residuals are heavier-tailed than Gaussian, so a Student-t variant remains an open decision.

5. Result: searching for short rules

A WRL program describes one neuron, and all 302 neurons run it in lockstep on the real wiring. To compare fairly, each program is converted into the same kernels as the baselines. It is simulated with and without a small pulse into each target, and the difference is read at the 12 kernel lags (a declared linear-response approximation). The program's few parameters are fitted by gradient descent through the JAX simulator, on training animals only. Because kernels come from the graph, a program also predicts pairs never seen in training.

Structural gates removed 296 of the first 400 enumerated programs because they never read the stimulus, and 76 because they cannot reach other neurons. 28 survived. The searches fitted the first 8 of those, plus hand-written rules, against B0, B1 and B3, on LOAO outer folds 0–2.

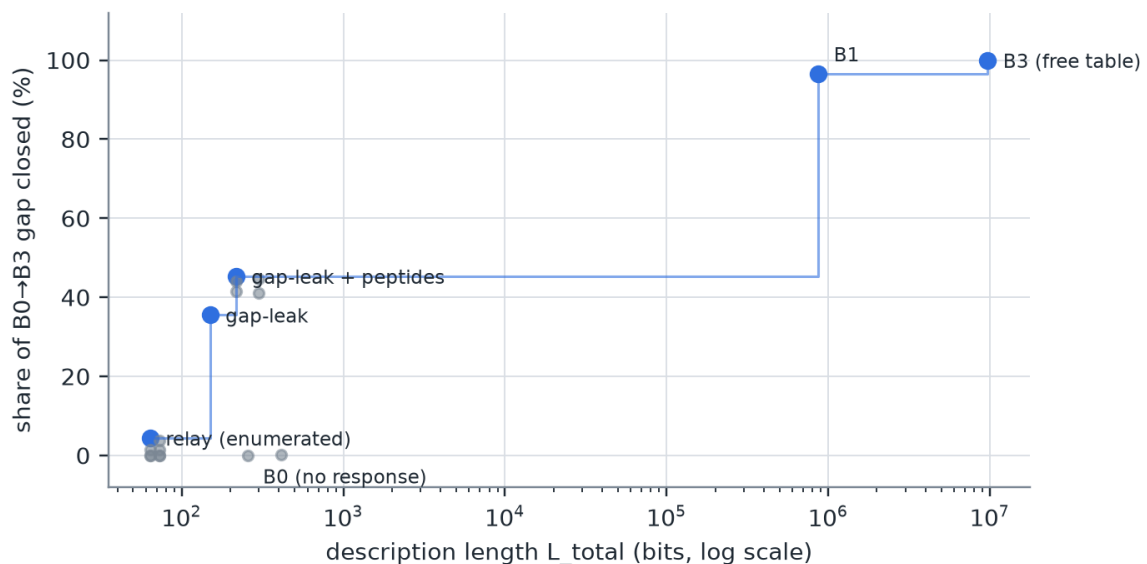


Figure 2. Size against accuracy, outer fold 0 (neuropeptide run). Accuracy is the share of the inner-validation improvement from B0 to B3. Blue points form the Pareto front: no other candidate is both shorter and more accurate.

Rule	What it does	Bits	Share of B0→B3 gap (folds 0/1/2)
relay (enumerated, b59c...)	next state = excitatory input + stimulus; no memory, no parameters	64	4% / 4% / 3%
gap-leak (hand-written)	leaky integrator of stimulus + signed chemical input, plus gap-junction coupling; 2 parameters	150	36% / 32% / 26%
gap-leak + peptides	as above, plus Ripoll-Sanchez 2023 short-range peptide routes with a fitted gain	217	45% / 41% / 32%
B1	one shape per target	~870k	96% / 97% / 97%

What this says

- **The free table still wins.** B3 is the frozen winner on every fold and is scored once on the held-out animal. No short rule matches it yet.
- **Short rules recover a real share at tiny size.** gap-leak is about 6,000 times shorter than B1 and sits on the Pareto front in every fold. Even the parameter-free relay beats “nothing happens”.

- **Its fitted time constant hits the top of its range.** On fold 0 gap-leak fits $\tau \approx 10$ s, at the upper bound. That matches the slow calcium-indicator kinetics measured earlier, so part of the rule's “memory” may stand in for the indicator. This is an interpretation, not yet a result.

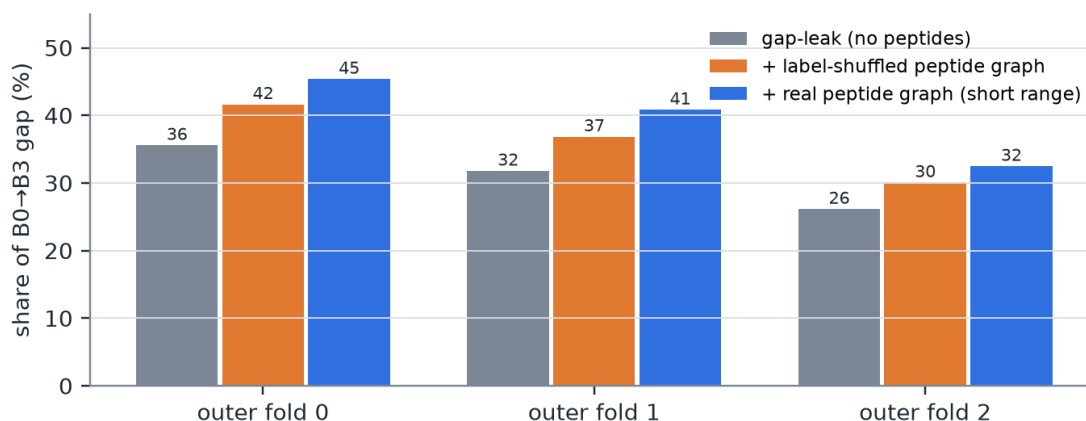


Figure 3. Adding neuropeptide routes to gap-leak. The real peptide graph beats an equally sized label-shuffled graph on all three folds. Most of the gain, though, comes from extra connectivity of any kind.

Neuropeptides. Short-range peptide routes raise gap-leak from 36/32/26% to 45/41/32%. A label-shuffled peptide graph of the same size reaches 42/37/30%. The real routes are better on every fold, but only by 2–4 points, and only one shuffle has been run. A proper null needs many shuffles and more folds.

6. Problems found and fixed

Stimulation-light artifact

An indicator fit came out degenerate. Tracing it showed that the stimulation light contaminates the imaging volume just before, at and just after each pulse. Those three volumes are now masked from all scoring (audit v2). Target coverage and the frozen split were unchanged.

Biased ridge selection

The first benchmark scored every candidate ridge penalty with one noise model, fitted at the middle penalty. That favoured whichever penalty matched it. Every kernel family chose the largest penalty on every fold, and B4 the smallest. The rule search exposed it: inner validation claimed kernels were *worse* than no response, while the held-out animal said the opposite. Each penalty now gets its own noise model, the grid was widened to 0.01–1000, and a regression test fails on the old code. In the rerun, no kernel family selects a grid edge. The affected partial runs were discarded.

A statistic that is sensitive to settings

“Share of the B0→B3 gap” is computed on inner-validation NLL. Fitting the inner noise models on a 10k-trace subsample instead of 30k (a speed change) left the rankings, the Pareto front and every held-out score unchanged. It did move gap-leak's share from 46/49/36% to 36/32/26%. Treat these shares as descriptive and compare them only within one run. Confirmatory claims will rest on paired held-out comparisons.

7. Limitations

- **Results are partial.** The benchmark covers T2a without the history term only; T2a with history and both T2b configurations are not yet run. The searches cover 3 of 113 folds, with very small budgets.
- **Many synapse signs are unknown.** Fenyves 2020 predicts signs for 1,287 excitatory and 406 inhibitory connections; 1,978 are unknown and are treated as excitatory.
- **Linear response.** WRL programs are judged by their response to a small pulse around rest, so saturation and threshold effects are invisible to the score.

- **Licensing.** The atlas data declares no license. This report contains only aggregate statistics; no raw or derived data is redistributed.

8. Next steps

- **Finish the benchmark** (about 15 hours on a plugged-in machine that stays awake). It resumes fold by fold from the pinned code. Then run the indicator false-sharing control, and freeze and publish the report. That closes OW-006/007.
- **Scale the search** to all outer folds and larger enumeration budgets, with budget curves and random-order controls. Add richer grammars: adaptation, saturation, multiple registers.
- **Make the neuropeptide test rigorous** with many label shuffles, so the real-graph gain can be compared against a null distribution.

Sources: artifacts/baselines-v1 (code f0de9f4, T2a-nohist complete); artifacts/search-atlas-demo (c5c19c9); artifacts/search-atlas-demo-np (41f4c6c). Each run has a run.json with input hashes.